

Invariant Mass as Qubit Mixedness and Null-Direction Distinguishability

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Abstract

For a finite family of lightlike momenta written as rank-one spinor matrices, $P = \sum_i \psi_i \psi_i^\dagger$, the invariant mass is $\mu = \sqrt{\det P}$. We prove, with every finite statement checked by the Lean 4 kernel, that this Lorentzian invariant is exactly a quantum information quantity of the normalized momentum matrix $\rho = P/\text{tr } P$: the squared mass-to-trace ratio is the qubit mixedness, $\mu^2/(\text{tr } P)^2 = \det \rho = (1 - \text{tr } \rho^2)/2$; for two spinors the rest gap is the energy-weighted trace distance of the normalized null directions, $\mu^2 = \|\psi\|^2 \|\phi\|^2 - |\langle \psi, \phi \rangle|^2$, the optimal one-shot distinguishability; and for arbitrary families the mass is an ensemble-weighted sum of pairwise distinguishability combinations. In the Pauli convention $P = E \mathbf{1} + \mathbf{p} \cdot \boldsymbol{\sigma}$ this reads $m^2/E^2 = 2(1 - \text{tr } \rho^2)$: the relativistic rest content of a momentum ensemble is precisely the purity deficit of its direction qubit. The oriented Pluecker coordinate $z = \psi \wedge \phi$ refines the picture: its modulus is the mixedness datum above, while its phase lives in the determinant $U(1)$ of the purification freedom $P = MM^\dagger$, $M \mapsto MR$. At fixed P , that phase is invisible to observables built only from the reduced momentum matrix and to the one-particle rest spectrum; the companion papers prove exactly which varying-phase and two-particle settings make it visible. “Mass is entropy” would overstate the result; the exact statement concerns purity, determinant, and pairwise distinguishability at fixed normalization, and nothing here predicts an absolute mass scale.

1 Setup and the purity identity

Let $\psi_1, \dots, \psi_N \in \mathbb{C}^2$ and $P = \sum_i \psi_i \psi_i^\dagger$. Under the Pauli map each summand is a future null momentum; $\det P$ is the Lorentzian invariant μ^2 , and the classical Cauchy–Binet identity $\det P = \sum_{i < j} |\psi_i \wedge \psi_j|^2$ is kernel-checked in the companion derivation paper [1]. Here we normalize: $T = \text{tr } P$ and, for $T \neq 0$, $\rho = P/T$, a positive unit-trace 2×2 matrix — literally a qubit density matrix built from the occupied null directions with energy weights.

Theorem 1 (Purity–determinant bridge). *For any 2×2 complex matrix ρ of unit trace,*

$$\det \rho = \frac{1 - \text{tr}(\rho^2)}{2}. \quad (1)$$

Consequently, for $T \neq 0$,

$$\frac{\mu^2}{T^2} = \det \rho = \frac{1 - \text{tr } \rho^2}{2}. \quad (2)$$

KERNEL (`det_eq_half_one_sub_purity_normalized_det.`) A single occupied direction, or several perfectly aligned directions, give a pure state ($\text{tr } \rho^2 = 1$) and zero mass; noncollinear directions give a mixed state and a positive rest gap; in the rest frame ρ is maximally mixed.

Corollary 1 (Pauli form). *Writing $P = E \mathbf{1} + \mathbf{p} \cdot \boldsymbol{\sigma}$ with real coefficients, $\text{tr } P = 2E$ and $\det P = E^2 - |\mathbf{p}|^2$, so for $E \neq 0$,*

$$\frac{m^2}{E^2} = 2(1 - \text{tr } \rho^2), \quad \rho = \frac{P}{2E}. \quad (3)$$

KERNEL (`pauli_trace_and_det`; the displayed convention is fixed by explicit Pauli matrices in the artifact.)

2 Mass as optimal distinguishability

Theorem 2 (Lagrange/trace-distance form). *For $\psi, \phi \in \mathbb{C}^2$,*

$$|\psi \wedge \phi|^2 = \|\psi\|^2 \|\phi\|^2 - |\langle \psi, \phi \rangle|^2. \quad (4)$$

KERNEL (`wedge_normSq_eq`.) For unit spinors the right side is $1 - |\langle \hat{\psi}, \hat{\phi} \rangle|^2$, the squared trace distance of the corresponding pure qubit states — the optimal one-shot discrimination probability bias. With the null normalization $\|\psi\|^2 = 2E_1$, $\|\phi\|^2 = 2E_2$,

$$\mu = 2\sqrt{E_1 E_2} D(\hat{\psi}, \hat{\phi}) : \quad (5)$$

after energy weighting, *rest mass is how well one measurement can tell the two null directions apart.*

Theorem 3 (Ensemble form). *For any finite family,*

$$\det P = \sum_{i < j} \left(\|\psi_i\|^2 \|\psi_j\|^2 - |\langle \psi_i, \psi_j \rangle|^2 \right), \quad (6)$$

so with $p_i = \|\psi_i\|^2 / T$,

$$\frac{\mu^2}{T^2} = \sum_{i < j} p_i p_j \left(1 - |\langle \hat{\psi}_i, \hat{\psi}_j \rangle|^2 \right). \quad (7)$$

KERNEL (`famMomentum_det_eq_pairwise`.) Invariant mass squared is an ensemble-weighted sum of pairwise quantum distinguishabilities. The maximally mixed witness (two orthogonal equal-energy directions: $\text{tr } \rho^2 = 1/2$, $\mu^2/T^2 = 1/4$) and the pure control (repeated direction: $\text{tr } \rho^2 = 1$, $\mu = 0$) are part of the artifact (`witness_and_control`).

3 Where the phase lives, and what this paper does not claim

Factor $P = MM^\dagger$ with $M = [\psi \ \phi]$; then $z = \det M = \psi \wedge \phi$ and the normalized amplitude $A = M/\sqrt{T}$ purifies $\rho = AA^\dagger$ with $\det A = z/T$. Right multiplication $M \mapsto MR$, $R \in U(2)$, preserves P while $z \mapsto z \det R$: *the modulus of z is the observable mixedness datum of Theorems 1–3, while its phase lives in the determinant $U(1)$ of the purification freedom.* At fixed P , every observable built only from the reduced momentum matrix is blind to that purification phase, and exact unitary conjugacy preserves the one-particle rest spectrum. The companion papers establish exactly which additional settings retain it: a spatially varying phase leaves derived link data with an integer winding, and a two-particle interference experiment separates equal-modulus fields with survival probability $4/5$ versus 1 [1].

Three boundaries. First, “mass is entropy” overstates the theorem: the exact statements concern the determinant, purity, and pairwise distinguishability at fixed trace normalization, not a von

Neumann entropy production principle. Second, nothing here selects an absolute scale: the identities are homogeneous, and the companion papers prove the natural positive homogeneous action selects only the collapsed zero scale. Third, Pluecker coordinates appear widely in quantum-information treatments of entanglement; the contribution claimed here is the exact, kernel-checked bridge from null-momentum ensembles to Lorentzian mass, the canonical rest operator, and operational discrimination — not the use of Pluecker coordinates in quantum information as such.

A Machine verification

All displayed identities are kernel-checked in Lean 4 over Mathlib in the module `PhysicsSM/Draft/NullEdge/MassMi` with build-enforced assumption footprints equal to `{propext, Classical.choice, Quot.sound}`, and integrate with the project’s existing trace-identity and visibility-duality modules. The archival release is a shared gate with the companion papers.

References

- [1] Null-Edge Formalization Project, *Null-spinor area as the rest gap of an exactly unitary Dirac walk*, companion paper, in preparation (2026).